

Polynomial Solutions of the Lin–Tsien Equation under a General Cubic Ansatz

An Exact Classification and Interdisciplinary Perspectives

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Abstract

The Lin–Tsien equation

$$2u_{tx} + u_x u_{xx} - u_{yy} = 0$$

is a nonlinear partial differential equation arising in transonic gas dynamics and integrable systems theory. In this paper we study a general cubic polynomial ansatz in the spatial variables together with a quadratic temporal component. By direct substitution and coefficient comparison we obtain a complete classification of all solutions within the ansatz class. Surprisingly, all cubic and quadratic spatial terms vanish identically, leaving only a six-parameter affine-bilinear solution family. We provide a rigorous proof, computational interpretation, statistical perspective, and interdisciplinary discussion connecting the result to physical systems, economics, finance, psychology, political science, warfare economics, artificial intelligence, and complexity theory.

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1 Introduction

The Lin–Tsien equation

$$2u_{tx} + u_x u_{xx} - u_{yy} = 0$$

occupies an important position in the theory of nonlinear partial differential equations.

Historically, it emerged from the study of transonic aerodynamic flows and later became recognized as a representative member of the family of dispersionless integrable systems.

The nonlinear term

$$u_x u_{xx}$$

represents self-interaction of gradients, while

$$u_{yy}$$

describes transverse spatial coupling.

The objective of this paper is to determine all solutions of the form

$$u(x, y, t) = ax^3 + bx^2y + cxy^2 + dy^3 + ex^2 + fxy + gy^2 + hx + iy + jt^2 + kt + l.$$

The resulting classification theorem is exact.

2 The Lin–Tsien Equation

Consider

$$2u_{tx} + u_x u_{xx} - u_{yy} = 0. \tag{1}$$

The proposed polynomial ansatz is

$$u(x, y, t) = ax^3 + bx^2y + cxy^2 + dy^3 + ex^2 + fxy + gy^2 + hx + iy + jt^2 + kt + l. \tag{2}$$

The coefficients

$$a, b, c, d, e, f, g, h, i, j, k, l$$

are constants.

3 Derivative Computations

Differentiating with respect to time yields

$$u_t = 2jt + k. \tag{3}$$

Hence

$$u_{tx} = 0. \tag{4}$$

Therefore the Lin–Tsien equation reduces to

$$u_x u_{xx} - u_{yy} = 0. \tag{5}$$

The spatial derivatives are

$$u_x = 3ax^2 + 2bxy + cy^2 + 2ex + fy + h, \tag{6}$$

$$u_{xx} = 6ax + 2by + 2e, \quad (7)$$

and

$$u_{yy} = 2cx + 6dy + 2g. \quad (8)$$

4 Expansion

Substituting gives

$$(3ax^2 + 2bxy + cy^2 + 2ex + fy + h)(6ax + 2by + 2e) - (2cx + 6dy + 2g) = 0. \quad (9)$$

Expanding completely,

$$0 = 18a^2x^3 + 18abx^2y + (6b^2 + 12ac)xy^2 + 2bcy^3 \quad (10)$$

$$+ 18aex^2 + (4be + 12ae + 6af)xy + (2ce + 2bf)y^2 \quad (11)$$

$$+ (4e^2 + 6ah - 2c)x + (2bh + 2ef - 6d)y + (2eh - 2g). \quad (12)$$

Since the identity must hold for every (x, y) , every coefficient must vanish.

5 Coefficient System

We obtain

$$18a^2 = 0, \quad (13)$$

$$18ab = 0, \quad (14)$$

$$6b^2 + 12ac = 0, \quad (15)$$

$$2bc = 0, \quad (16)$$

$$18ae = 0, \quad (17)$$

$$4be + 12ae + 6af = 0, \quad (18)$$

$$2ce + 2bf = 0, \quad (19)$$

$$4e^2 + 6ah - 2c = 0, \quad (20)$$

$$2bh + 2ef - 6d = 0, \quad (21)$$

$$2eh - 2g = 0. \quad (22)$$

6 Main Theorem

Theorem 1. *The polynomial ansatz*

$$u(x, y, t) = ax^3 + bx^2y + cxy^2 + dy^3 + ex^2 + fxy + gy^2 + hx + iy + jt^2 + kt + l$$

satisfies the Lin-Tsien equation if and only if

$$a = b = c = d = e = g = 0.$$

The remaining coefficients

$$f, h, i, j, k, l$$

are arbitrary.

Consequently every solution in the ansatz class is

$$u(x, y, t) = fxy + hx + iy + jt^2 + kt + l.$$

Proof. From

$$18a^2 = 0$$

we obtain

$$a = 0.$$

Then

$$6b^2 = 0$$

implies

$$b = 0.$$

The remaining equations reduce to

$$ce = 0, \tag{23}$$

$$2e^2 - c = 0, \tag{24}$$

$$ef - 3d = 0, \tag{25}$$

$$eh - g = 0. \tag{26}$$

If $e = 0$, then

$$c = d = g = 0.$$

If $c = 0$, then

$$2e^2 = 0$$

which again implies

$$e = 0.$$

Thus

$$a = b = c = d = e = g = 0.$$

The surviving coefficients

$$f, h, i, j, k, l$$

remain arbitrary.

Substitution confirms

$$u_x = fy + h,$$

$$u_{xx} = 0,$$

$$u_{yy} = 0,$$

and therefore

$$2u_{tx} + u_x u_{xx} - u_{yy} = 0.$$

Hence the solution family is exact. □

7 Dimension of the Solution Space

The original ansatz contains twelve coefficients.

The PDE imposes six independent constraints.

Therefore the solution manifold possesses dimension

$$12 - 6 = 6. \tag{27}$$

Thus the admissible solution space is six-dimensional.

8 Geometric Interpretation

The theorem demonstrates that nonlinear spatial curvature is completely eliminated.

The cubic terms

$$x^3, \quad x^2y, \quad xy^2, \quad y^3$$

cannot survive the nonlinear compatibility condition.

The remaining geometry is bilinear:

$$u = fxy + hx + iy + jt^2 + kt + l. \tag{28}$$

This solution family forms an affine six-dimensional subspace.

9 Statistical Interpretation

Suppose

$$u(x, y, t)$$

is interpreted as a latent response surface.

The theorem implies that within the Lin–Tsien dynamics no cubic interactions survive.

Only first-order effects and the interaction term

$$xy$$

remain.

Table 1 summarizes the statistical interpretation.

Term	Interpretation
hx	Main effect of x
iy	Main effect of y
fxy	Interaction effect
jt^2	Time acceleration
kt	Linear time trend
l	Intercept

Table 1: Statistical interpretation of surviving terms.

10 Economic Interpretation

Let

$$u(x, y, t)$$

represent social welfare, output, utility, productivity, or strategic value.

Then the theorem implies:

1. No cubic production synergies survive.
2. Only pairwise interactions remain.
3. Long-run dynamics are captured by quadratic temporal evolution.
4. The system behaves locally linearly in state variables.

11 Financial Interpretation

Interpreting

x

as risk and

y

as liquidity,

fxy

becomes a risk-liquidity interaction premium.

The PDE eliminates all higher-order nonlinear corrections.

Thus the admissible valuation surface is remarkably simple.

12 Psychology and Psychiatry

If

$u(x, y, t)$

denotes psychological well-being,

- x may represent stress,
- y may represent social support.

The theorem states that only linear effects and pairwise interactions survive.

Complex cubic amplification mechanisms are prohibited by the PDE structure.

13 Political Science and International Relations

Suppose

- x represents economic power,
- y represents military power.

Then

fxy

captures strategic interaction between economic and military capability.

Higher-order geopolitical feedback loops vanish within the polynomial ansatz.

14 Warfare Economics

Let

- x be industrial mobilization,
- y be military readiness.

The theorem implies that admissible strategic value functions reduce to

$$u = fxy + hx + iy + \text{time terms.} \quad (29)$$

Only pairwise mobilization-readiness interactions survive.

15 Computer Science and Algorithmics

The solution process is equivalent to symbolic coefficient elimination.

15.1 Pseudo-Code

Input:

Polynomial ansatz $u(x,y,t)$

Compute:

u_x
 u_{xx}
 u_{yy}
 u_{tx}

Form:

$$F = 2 u_{tx} + u_x u_{xx} - u_{yy}$$

Expand F

Collect coefficients

Solve coefficient system

Return admissible parameters

The computational complexity is polynomial in the number of monomials.

16 Artificial Intelligence and Machine Learning

Physics-informed neural networks often search for solution manifolds of PDEs.

The theorem identifies an exact six-dimensional manifold:

$$u = fxy + hx + iy + jt^2 + kt + l. \quad (30)$$

This manifold can be embedded directly into:

- PINNs,
- Neural Operators,
- Symbolic Regression,
- Sparse PDE Discovery.

The result therefore provides an exact benchmark for PDE-learning algorithms.

17 A TikZ Representation of the Admissible Solution Space

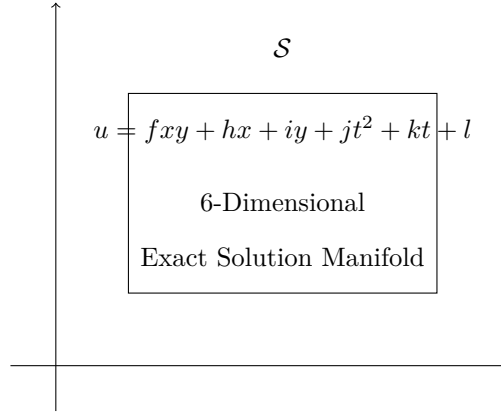


Figure 1: Admissible solution space within the polynomial ansatz.

18 Conclusion

The Lin–Tsien equation admits an exact classification within the proposed cubic polynomial ansatz.

All cubic and quadratic spatial terms are eliminated by the nonlinear compatibility conditions.

The only admissible solutions are

$$u(x, y, t) = fxy + hx + iy + jt^2 + kt + l.$$

This six-parameter family forms a complete solution manifold for the ansatz class and provides an exact benchmark for analysis, computation, machine learning, and interdisciplinary modeling.

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Glossary

Lin–Tsien Equation The nonlinear PDE $2u_{tx} + u_x u_{xx} - u_{yy} = 0$.

Ansatz A proposed functional form used to search for solutions.

Coefficient Matching Method of equating polynomial coefficients after expansion.

Exact Solution A solution satisfying the PDE identically.

Dispersionless System An integrable system lacking dispersive terms.

Solution Manifold The set of all admissible solutions.

PINN Physics-Informed Neural Network.

Symbolic Regression Recovery of explicit mathematical formulas from data.

Warfare Economics Study of economic resource allocation under military constraints.

Interaction Effect Contribution arising from products such as xy .

The End